

Measurement-Based Quantum Computation

A synopsis of Jozsa's *An Introduction to Measurement Based Quantum Computation*

Sanjay Suresh

June 2026

Abstract

In the circuit model, computation is unitary evolution and measurement happens only at the end. Measurement-based quantum computation (MBQC) inverts this: one begins with a large fixed entangled state and computes by measuring qubits one at a time, choosing later measurement bases according to earlier outcomes. Two models realize this idea — teleportation quantum computation (TQC) and the one-way / cluster-state quantum computer (1WQC). These notes synthesize R. Jozsa's introduction ([quant-ph/0508124](https://arxiv.org/abs/quant-ph/0508124)): how each model works, why each is universal, the three senses in which they are the same model, and what MBQC reveals about computational complexity.

1 Notation

Pauli operators I, X, Z with $iY = ZX$, and $\oplus$ is addition modulo 2. The two-qubit gates are

$$CX \, |i\rangle |j\rangle = |i\rangle |i \oplus j\rangle, \quad CZ \, |i\rangle |j\rangle = (-1)^{ij} |i\rangle |j\rangle,$$

with $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Unlike CX , CZ is symmetric in its two inputs. The Bell basis is generated from $|B_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ by Paulis on one half,

$$|B_{cd}\rangle = (Z^c X^d \otimes I) |B_{00}\rangle,$$

and a useful maximally entangled state mixing the standard and Hadamard bases is

$$|H\rangle = \frac{1}{\sqrt{2}}(|0\rangle |+\rangle + |1\rangle |-\rangle) = CZ \, |+\rangle |+\rangle.$$

2 Gate teleportation: the heart of TQC

Standard teleportation sends $|\alpha\rangle$ from qubit 1 to qubit 3 using a Bell pair on 2–3 and a Bell measurement on 1–2; outcome $|B_{cd}\rangle$ leaves qubit 3 in $X^d Z^c |\alpha\rangle$. Gottesman and Chuang's insight is to measure in a *rotated* Bell basis. For any one-qubit gate U , define

$$|B(U)_{cd}\rangle = (U^\dagger \otimes I) |B_{cd}\rangle.$$

Measuring 1–2 in this basis teleports *and* applies the gate at once: qubit 3 is left in $X^d Z^c U |\psi\rangle$. A gate is realized purely by a choice of measurement.

The clean statement: for any maximally entangled $|\phi\rangle = \frac{1}{\sqrt{d}} \sum_i |i\rangle |i\rangle$ in dimension d , projecting $|\alpha\rangle_1 |\phi\rangle_{23}$ onto $|\phi\rangle_{12}$ yields $\frac{1}{d} |\alpha\rangle_3$.

Lemma 1. *The projection of $|\alpha\rangle_1 |\phi\rangle_{23}$ onto $|\phi\rangle_{12}$ leaves qubit 3 in $\frac{1}{d} |\alpha\rangle_3$.*

Replacing $|\phi\rangle$ by the rotated state $|\phi(U)\rangle = (U^\dagger \otimes I) |\phi\rangle$ gives $\frac{1}{d} U |\alpha\rangle_3$.

Lemma 2. *The projection of $|\alpha\rangle_1 |\phi\rangle_{23}$ onto $|\phi(U)\rangle_{12}$ leaves qubit 3 in $\frac{1}{d} U |\alpha\rangle_3$.*

Reading these projections as outcomes of a measurement means choosing d^2 unitaries U_i so that $\{|\phi(U_i)\rangle\}$ is an orthonormal basis — equivalently $\{U_i\}$ is a *unitary operator basis*, $\text{Tr}(U_i U_j^\dagger) = d \delta_{ij}$. One may go further and use $n \geq d^2$ operators forming a POVM,

$$\sum_{i=1}^n k_i |\phi(U_i)\rangle \langle \phi(U_i)| = I_d \otimes I_d,$$

so the Bell measurement can even be replaced by a (generally non-projective) POVM. Two-qubit gates such as CZ come from the same lemmas in dimension $d = 4$ (a 16-dimensional measurement), or, more cleverly, from smaller measurements on a small entangled gadget.

3 Byproducts and adaptive measurements

The catch: teleporting a gate never gives $U |\psi\rangle$ exactly, but $PU |\psi\rangle$ with a random Pauli P fixed by the outcome. Chaining gates $\dots U_3 U_2 U_1 |\psi\rangle$ produces $\dots P_3 U_3 P_2 U_2 P_1 U_1 |\psi\rangle$ — Paulis interleaved everywhere. The fix is to commute the Paulis to the end and absorb the damage by adapting later measurement bases.

For rotations $R_x(\theta) = e^{-i\theta X}$, $R_z(\theta) = e^{-i\theta Z}$ the propagation rules are

$$\begin{aligned} R_x(\theta)X &= X R_x(\theta), & R_x(\theta)Z &= Z R_x(-\theta), \\ R_z(\theta)X &= X R_z(-\theta), & R_z(\theta)Z &= Z R_z(\theta). \end{aligned}$$

Pushing a byproduct $Z^a X^b$ past the next rotation only flips the sign of its angle (by $(-1)^b$). So if the previous outcome was b , one measures for $R_z((-1)^b \beta)$ instead of $R_z(\beta)$ — an *adaptive* basis choice that cancels the byproduct and reinstates the intended gate. Iterating, the computation yields

$$\dots X_2^{m_2} Z_2^{n_2} X_1^{m_1} Z_1^{n_1} U |\psi\rangle,$$

with exponents m_i, n_i bit-sums (mod 2) of measurement outcomes. The final readout is a Z -basis measurement; a leftover X^m just flips the interpretation of that bit ($k_i \mapsto k_i \oplus m_i$) and a leftover Z^n does nothing. A striking consequence: these final Z measurements are never adaptive, so they may be performed *first* — the output can be measured before the computation that determines its meaning has run.

4 Clifford gates parallelize

The Pauli group $\mathcal{P}_n$ is generated by tensor products of $\pm I, \pm iI, X, Z$. A unitary C is a *Clifford operation* if it normalizes the Paulis,

$$C \mathcal{P}_n C^\dagger = \mathcal{P}_n,$$

i.e. Paulis propagate across C while C itself stays fixed — no adaptation needed. CZ, H ($HX = ZH, HZ = XH$) and the $\pi/4$ phase gate are Clifford, and the whole Clifford group on n qubits is generated by $Z, H, P_{\pi/4}, CX$. Since no measurement depends on another, an entire array of

Clifford gates can be done in *one parallel layer of measurements*; the byproduct exponents are then computed as bit-sums, a purely classical $O(\log)$ -depth task.

This sharpens the Knill–Gottesman theorem (Clifford circuits are classically simulable in polynomial time, even in log depth): MBQC shows the genuinely quantum content of a Clifford array can be *compressed into constant quantum depth* plus classical post-processing — a structural statement the classical-simulation result does not provide.

5 The one-way quantum computer

The 1WQC starts from a *cluster state*: lay qubits in $|+\rangle$ on a 2D grid and apply CZ to every neighbouring pair. All the CZ s commute, so the resource is built in *constant* quantum depth regardless of size. Computation uses only single-qubit measurements in the bases

$$M_z = \{|0\rangle, |1\rangle\}, \quad M(\theta) = \{|0\rangle \pm e^{i\theta} |1\rangle\},$$

($\theta = 0$ is an X -measurement); outcomes are uniformly random. As in TQC, gates appear only up to Pauli corrections $X^a Z^b$, so bases are again adaptive. The name “one-way” reflects that the pristine cluster is irreversibly consumed as the layers of measurement proceed.

A single-qubit $U = R_x(\zeta)R_z(\eta)R_x(\xi)$ is implemented on a five-qubit line: place $|\psi\rangle$, append four $|+\rangle$, entangle, and measure qubits 1–4 adaptively in bases set by the Euler angles. The output qubit 5 holds $X^{s_2+s_4}Z^{s_1+s_3}U|\psi\rangle$. The atomic step is one $M(\theta)$ measurement, which realizes the gate

$$W(\theta) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & e^{i\theta} \\ 1 & -e^{i\theta} \end{pmatrix} = H P(\theta), \quad P(\theta) = \text{diag}(1, e^{i\theta}),$$

leaving the next qubit in $X^s W(-\theta)|\psi\rangle$. Since any one-qubit gate factors as $U = W(0)W(\theta_1)W(\theta_2)W(\theta_3)$, repeating this step along a wire builds any single-qubit unitary; $\{CZ, W(\theta)\}$ and $\{CZ, R_x(\theta), R_z(\theta)\}$ are universal. Input states can be eliminated by starting every wire in $|+\rangle$, so the whole resource is a standard cluster state. Two-qubit gates use the second grid dimension: an explicit pattern realizes CZ on a small 2D block, and CZ with single-qubit gates suffices for universality.

Z -measurements delete qubits

Because a general circuit maps onto an irregular region of a large rectangular cluster, unused sites must be removed. Measuring a site A in the Z basis (outcome k_A) deletes it from the graph: it disconnects A from its neighbours and leaves only an extra Z^{k_A} byproduct on each former neighbour. These deletions are non-adaptive, so they too can be done first, carving the desired graph out of the cluster before any real computation.

6 Why one-dimensional clusters fail

2D clusters are universal; 1D clusters are not. A 1D cluster state has the ladder structure of a matrix-product state, and any sequence of single-qubit measurements on it — even adaptive ones — can be classically simulated in polynomial time. The argument sweeps the chain holding only one or two sites’ reduced states at a time, applying a measurement projector at measured sites and a partial trace at unmeasured ones, computing each outcome probability

$$P(k' | a', b', \dots) = \frac{P(a', b', \dots, k', \dots)}{P(a', b', \dots)}$$

by passing through the ladder n times. If a 1D model were universal, quantum computation would collapse to classical polynomial time — so genuine speedup needs the 2D entanglement structure.

7 Three bridges between TQC and 1WQC

The models look different — TQC uses multi-qubit Bell measurements on bipartite pairs, 1WQC uses single-qubit measurements on a multipartite cluster — but they are three views of one idea.

1. **Pairing measurements (Aliferis–Leung).** Since $CX|B_{ij}\rangle = |(-1)^j\rangle|i\rangle$ (first ket in the $\pm$ basis), a Bell measurement factors into an entangling CX plus single-qubit measurements. Special *pairs* of consecutive 1WQC measurements fuse into one Bell measurement, recovering TQC.
2. **One-bit teleportation (Childs et al.; Jorrand–Perdrix).** Teleportation decomposes into two identical primitive steps “adjoin $|+\rangle$, entangle with CZ , measure one qubit.” Using the $|H\rangle$ resource with Bell basis $\{(P \otimes I)|H\rangle\}$ — which CZ maps to $\{|\pm\rangle|\pm\rangle\}$ — makes the two steps symmetric. This shared “one-bit teleportation” is the common atom of both models.
3. **Matrix-product / valence-bond states (Verstraete–Cirac).** Take a grid of $|H\rangle$ bonds and at each site project onto the two-dimensional code subspace with

$$\Pi = |\bar{0}\rangle\langle 0 \cdots 0| + |\bar{1}\rangle\langle 1 \cdots 1|.$$

The projected state is exactly the 1WQC cluster state, and a cluster-state $M(\theta)$ measurement is precisely a full teleportation “cut down” by Π : the rotated Bell states

$$|a=0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + e^{i\theta}|11\rangle), \quad |a=1\rangle = \frac{1}{\sqrt{2}}(|00\rangle - e^{i\theta}|11\rangle)$$

lie inside the Π -subspace and reproduce the $M(\theta) = \{|\bar{0}\rangle \pm e^{i\theta}|\bar{1}\rangle\}$ measurement. Each 1WQC ingredient is a descendant, under Π , of a teleportation on the valence-bond grid.

8 Complexity and the classical/quantum split

Both MBQC models are polynomial-time equivalent to the circuit model, with only linear overhead in either direction (a circuit measurement is simulated by an ancilla and a CX ; a 1WQC pattern is simulated by building the cluster and adding a basis-rotation gate per measurement). So they compute the same class of problems.

What changes is structure. MBQC exposes parallelism the circuit model hides: non-adaptive measurements commute, so Clifford arrays finish in one layer, and any polynomial-size circuit maps to a polynomial number of measurement layers. More deeply, MBQC gives a natural way to separate an algorithm into *quantum layers* (depth-1 bursts of measurement) and *classical layers* (the adaptive bookkeeping between them) — a separation with no analogue in classical complexity, and a fitting home for the heavy classical post-processing in algorithms like Shor’s. This motivates Jozsa’s conjecture.

Conjecture. *Any polynomial-time quantum algorithm can be implemented with only $O(\log n)$ quantum layers interspersed with polynomial-time classical computation.*

It would mean polynomial-time classical computation needs only a thin sliver of “quantum assistance” to reach the full power of quantum computation. It is open in general but known to hold for Shor’s algorithm (Cleve–Watrous).

Takeaways

- Universal computation is possible with measurements alone, on a fixed entangled resource.
- Randomness of outcomes is tamed by Pauli propagation and adaptive bases; what survives is a final Pauli that only relabels the answer.
- Clifford operations need no adaptation, so they parallelize to constant quantum depth.
- 2D cluster entanglement is essential — 1D is classically simulable.
- TQC and 1WQC are the same model seen through teleportation, one-bit teleportation, or matrix-product states.
- MBQC’s real payoff is conceptual: a clean quantum/classical separation and new questions about how little “quantum” a quantum algorithm really needs.

References

- [1] R. Jozsa, *An introduction to measurement based quantum computation*, arXiv:quant-ph/0508124 (2005).
- [2] D. Gottesman and I. Chuang, *Quantum teleportation as a universal computational primitive*, Nature **402**, 390–393 (1999); arXiv:quant-ph/9908010.
- [3] R. Raussendorf and H. J. Briegel, *A one-way quantum computer*, Phys. Rev. Lett. **86**, 5188–5191 (2001); arXiv:quant-ph/0010033.
- [4] R. Raussendorf, D. E. Browne, and H. J. Briegel, *Measurement-based quantum computation with cluster states*, Phys. Rev. A **68**, 022312 (2003); arXiv:quant-ph/0301052.
- [5] F. Verstraete and J. I. Cirac, *Valence bond solids for quantum computation*, Phys. Rev. A **70**, 060302(R) (2004); arXiv:quant-ph/0311130.